

# stellarrouter

The OpenRouter-style LLM gateway with **verifiable, on-chain billing** — AI agents pay per call in USDC, humans prepay into a refundable Soroban vault.

[stellarrouter.vercel.app](https://stellarrouter.vercel.app)

[github.com/Lexirieru/stellarrouter](https://github.com/Lexirieru/stellarrouter)

Live on Stellar testnet

## PROBLEM

# LLM access is built for credit cards, not for agents – or auditors

## Agents can't pay

Every LLM API hides behind subscriptions and human-owned keys. An autonomous agent cannot open a Stripe account — so it borrows its operator's key: unbounded spend, no per-request settlement, no machine-native way to buy one completion.

## Humans can't audit

Gateway credits are opaque custodial database rows. You can't verify what you were charged for, and your unused balance lives entirely at the provider's mercy.

## SOLUTION

# One gateway, two payment doors

### Agent door · x402

Request → `402 Payment Required` → pay **\$0.005 USDC** → completion. No account, no API key, and **zero XLM** — network fees are sponsored by the OpenZeppelin Channels facilitator. Settles on-chain in ~5s.

### Human door · prepaid

OpenRouter ergonomics — catalog, keys, logs — but the balance lives in the `credits` Soroban contract. Every deposit/debit/withdraw emits a public event (**proof-of-billing**), and `withdraw()` refunds unused USDC any time.

**The differentiator:** billing that OpenRouter structurally cannot offer — non-custodial, refundable, publicly auditable.

# The only chain where \$0.005-per-call works today

## Native USDC (SAC / SEP-41)

~5s finality, sub-cent fees — micro-billing granularity that card rails and L1 gas can't touch.

## x402 + OZ Channels (official stack)

Fee sponsorship means an agent needs **only USDC, zero XLM** — the official Stellar agentic-payments rail, in production.

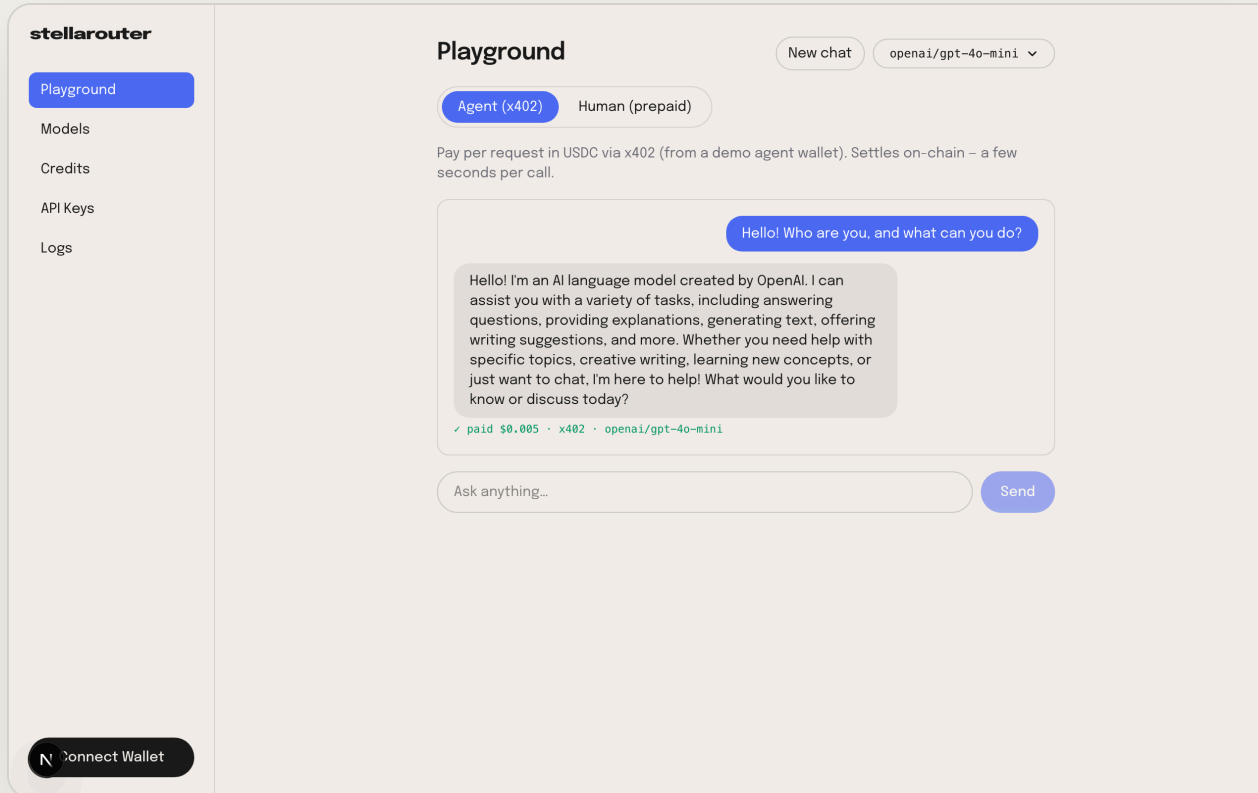
## Soroban events = proof-of-billing

`require_auth` custody boundary: the admin can only debit usage; users always exit with their balance.

## Anchors (SEP-24) next

Fiat on-ramp into credits — LLM access for markets without cards.

# A complete console, live on testnet



Playground with **agent (x402)** and **human (prepaid)** modes — a real paid receipt on screen.

Plus: 560-model catalog with network-aware availability, wallet-bound API keys (SEP-10-style challenge), usage logs, live on-chain activity feed, analytics, in-product feedback.

Multi-wallet (Stellar Wallets Kit)

Mobile responsive

29 tests

# Real usage, verifiable on the explorer

54

user wallets onboarded

280+

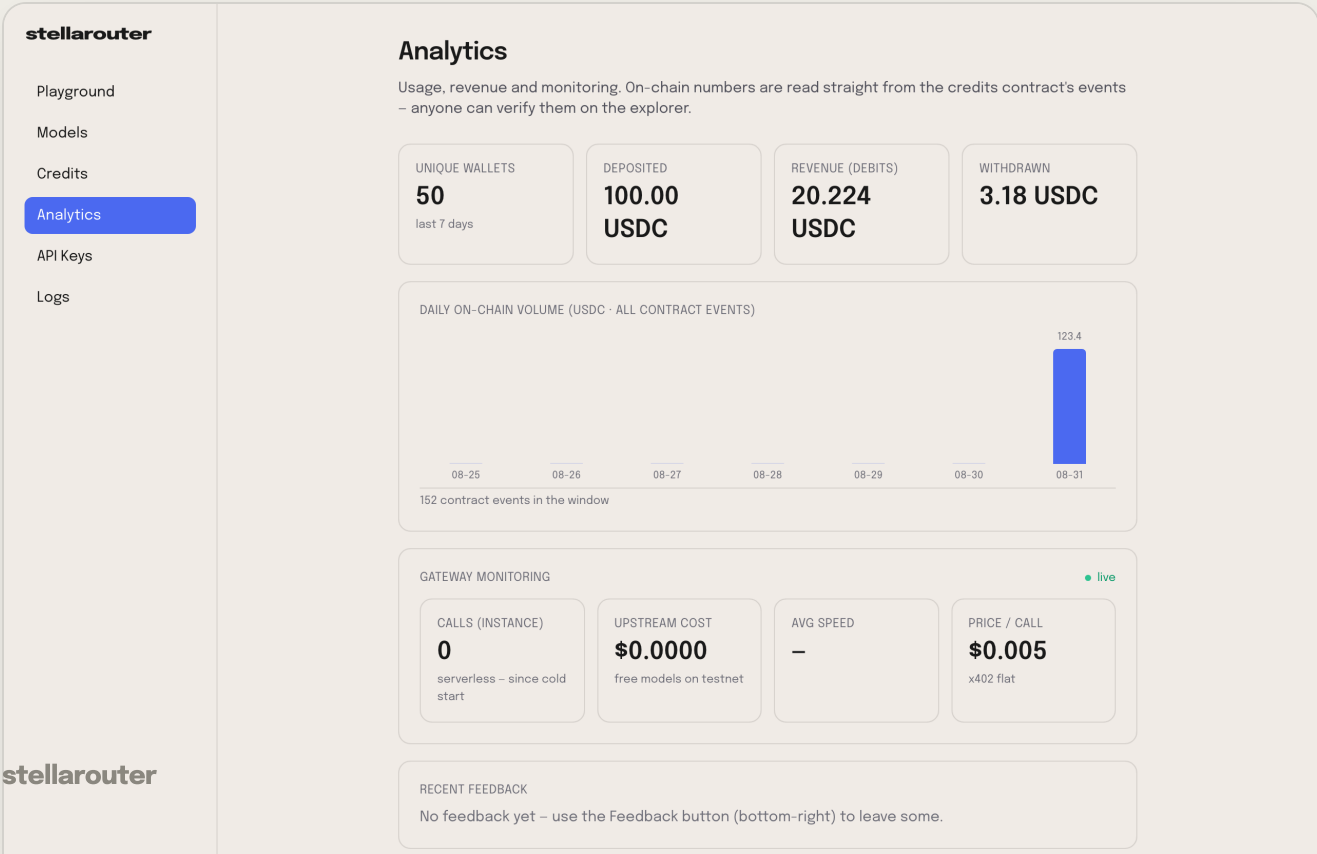
signed on-chain transactions

110

USDC deposited to the vault

21.8

USDC gateway revenue (debits)



Every number is read from contract events — the analytics page recomputes them client-side, and anyone can audit them on stellar.expert. Feedback loop closed: 10 pilot findings shipped, each traceable to a commit.

# The agentic-payments wave needs a product layer

**Demand:** LLM gateways already clear billions of tokens a day for humans — and the fastest-growing consumers (autonomous agents) are locked out of all of them.

**Ecosystem validation:** agentic payments is an official Stellar use case; 12+ x402 projects in the directory, REAPP raised **\$70k SCF** — all SDKs, middleware or marketplaces.

**The gap we take:** nobody ships the **product layer** — an OpenRouter-class gateway with catalog, keys, logs and on-chain billing. Stellarrouter is that product.

# Console → Gateway → Contract, all in production

## Next.js console

Vercel

Playground · Models · Credits · Keys ·  
Logs · Analytics  
Stellar Wallets Kit · RPC getEvents live  
feed



## Express gateway

Vercel serverless

x402 door (OZ Channels verify/settle)  
Prepaid door: hashed keys ↔ wallet,  
balance check → completion →  
`debit()`  
Network-aware model policy · /metrics  
· /feedback



## `credits` contract

Soroban · testnet

deposit / debit / withdraw / collect  
events on every state change  
cross-contract USDC SAC transfers  
9 unit tests

Upstream: any OpenAI-compatible provider · Mainnet = config flip (pubnet + OZ mainnet + Circle USDC), no code changes.

# Win the agents first – humans follow the models

## 1 • Agent-native onboarding

Drop-in templates (fetch/MCP/SKILL.md) so any framework pays Stellarrouter out of the box — the 402 *is* the signup.

## 2 • Ecosystem distribution

Stellar directory + agent-tool registries; convert Stellar Hacks: Agents & MCP builders into first integrators.

## 3 • Human retention

Refundable balances remove lock-in fear; proof-of-billing becomes the compliance-friendly wedge for teams.

## 4 • Anchor partnership

SEP-24 fiat top-ups → LLM access for card-less markets. KPIs: agents integrated · paid calls/day · USDC settled.

# From testnet MVP to the agent-payments default

## Now → Q3

Streaming (SSE) through both doors ·  
persistent hosted store · per-model x402  
pricing tiers

## Q4 2026

**Mainnet flip** (pubnet + OZ mainnet +  
Circle USDC) · contract audit · treasury  
ops · SEP-24 top-ups

## 2027

MPP payment channels for high-  
frequency agents · model-provider  
revenue share in USDC

**stellarrouter**

OpenRouter whose bills you can audit on the blockchain

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